

Dataset Overview

The Telco Customer Churn dataset contains customer information from ABC Communications. It includes customer demographics, subscribed services, billing information, payment methods, contract details, and customer churn status. The dataset was inspected to evaluate its quality before carrying out the business analysis.

Dataset Structure

- Number of Records (Rows): 7,043
- Number of Variables (Columns): 21

Each row represents one customer, while each column represents a specific customer attribute, such as demographics, subscribed services, payment information, or churn status.

Data Types

The dataset contains both categorical and numerical variables.

Categorical Variables

- CustomerID
- Gender
- Partner
- Dependents
- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies
- Contract
- PaperlessBilling

- PaymentMethod
- Churn

Numerical Variables

- SeniorCitizen
- Tenure
- MonthlyCharges
- TotalCharges

After inspection, all columns were found to have the appropriate data type for analysis.

Missing Values

The dataset was inspected for missing values using Excel's Go To Special feature.

No missing values were found in any column except TotalCharges, where 11 blank cells were identified.

Further investigation showed that all affected customers had a tenure of 0 months. Since the TotalCharges column represents the cumulative amount paid by a customer since joining the company, customers with zero months of tenure have not yet accumulated any charges. Therefore, the blank values are consistent with the business context rather than indicating random data entry errors.

Duplicate Records

The dataset was checked for duplicate records.

No duplicate records were found. Each customer record is unique and can be identified using the CustomerID column.

Feature Engineering

A new variable named **Charge Category** was created by grouping the *Monthly Charges* into three categories (**Low, Medium, and High**) to enable easier comparison of churn rates across different pricing levels.

Statistical Summary

The descriptive statistics provide an overview of the numerical variables in the dataset and reveal the overall characteristics of ABC Communication Ltd.'s customers.

- The dataset contains 7,043 customer records.
- The average customer tenure is 32.37 months, indicating that customers typically remain with the company for nearly three years.
- Customer tenure ranges from 0 to 72 months, showing that the dataset includes both new and long-term customers.
- The average monthly charge is 64.76, with values ranging from 18.25 to 118.75, indicating considerable variation in customer billing.
- The average total charge is 2,283.30, reflecting the cumulative amount customers have paid over their subscription period.
- The median tenure of 29 months suggests that half of the customers have stayed with the company for less than 29 months, while the other half have remained for longer.
- The presence of 11 blank values in the TotalCharges column was investigated and found to correspond to customers with zero tenure, meaning they had not yet accumulated any charges.